

HEQU, China

WMO: 535640

Lat: 39.3649N

Lon: 111.2130E

Elev: 1037

StdP: 89.47

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.8	-18.7	-27.9	0.3	-14.3	-25.7	0.4	-13.0	7.2	-6.8	6.1	-3.3	0.5	20	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.7	18.8	32.2	18.5	30.9	18.2	22.8	28.5	22.0	27.5	21.2	26.5	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	18.3	24.9	20.5	17.2	24.3	19.6	16.3	23.9	73.7	28.6	70.1	27.4	67.0	26.5	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
7.3	6.2	5.1	DB	-23.7	36.2	2.8	1.7	-25.8	37.4	-27.4	38.4	-29.0	39.4	-31.1	40.7	
			WB	-24.0	24.5	2.8	0.9	-26.0	25.1	-27.6	25.7	-29.2	26.2	-31.2	26.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-9.2	-3.6	4.1	12.0	17.8	22.3	24.2	22.1	16.7	9.6	1.2	-6.8
	DBStd	12.02	4.13	4.99	4.68	4.61	3.73	2.78	2.26	2.54	3.31	3.87	4.15	4.07
	HDD10.0	2041	595	382	191	31	1	0	0	0	1	54	265	522
	HDD18.3	3798	853	616	442	194	54	4	0	3	68	270	514	780
	CDD10.0	1772	0	0	7	92	243	369	439	376	202	42	0	0
	CDD18.3	487	0	0	0	5	38	123	181	120	19	1	0	0
	CDH23.3	5145	0	0	2	128	617	1387	1756	1010	234	11	0	0
	CDH26.7	1967	0	0	0	33	206	579	757	337	54	0	0	0
Wind	WSAvg	1.6	1.1	1.4	2.1	2.4	2.3	2.0	1.6	1.4	1.4	1.5	1.4	1.2
Precipitation	PrecAvg	400	3	4	11	18	34	50	105	87	54	22	10	2
	PrecMax	649	10	12	58	54	71	99	305	189	109	87	53	12
	PrecMin	236	0	0	0	0	2	6	34	10	13	2	0	0
	PrecStd	113	3	4	13	14	19	22	62	49	29	19	13	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.9	13.1	22.3	30.6	33.3	35.5	35.8	33.6	31.3	24.9	16.3	5.9
		MCWB	-0.5	4.2	9.3	13.5	15.3	17.5	20.9	20.6	17.1	14.5	6.9	-0.2
	2%	DB	2.7	9.9	18.5	26.8	31.0	33.5	34.0	31.8	28.0	22.4	13.3	3.6
		MCWB	-2.2	2.5	7.1	11.7	14.6	17.0	20.2	20.3	16.6	12.2	5.9	-1.2
	5%	DB	0.5	7.5	15.8	24.2	29.1	31.8	32.4	30.1	26.0	19.9	11.0	1.9
		MCWB	-3.6	1.0	5.6	10.7	13.7	16.8	20.0	20.1	16.1	11.2	4.8	-2.1
	10%	DB	-1.5	5.1	13.2	21.6	27.0	30.2	30.9	28.5	24.1	17.6	8.7	0.2
		MCWB	-4.9	-0.3	4.3	9.5	12.9	16.5	20.0	19.8	15.5	10.3	3.8	-3.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	5.0	10.0	15.1	17.6	21.3	24.2	23.7	20.9	15.5	9.2	0.9
		MCDB	4.5	11.6	20.2	24.2	27.3	28.3	29.5	28.9	25.2	22.6	12.2	4.0
	2%	WB	-1.8	3.3	7.8	13.5	16.2	20.0	23.2	22.7	19.1	13.7	7.3	-0.5
		MCDB	2.3	8.8	16.4	22.2	25.5	26.6	29.0	28.0	23.9	19.3	11.3	2.6
	5%	WB	-3.4	1.6	6.3	12.1	15.2	19.2	22.3	21.9	17.9	12.5	5.8	-1.7
		MCDB	0.3	6.5	14.4	21.5	24.5	26.2	28.2	26.7	22.8	17.9	9.7	1.6
	10%	WB	-4.8	0.1	4.8	10.8	14.3	18.3	21.6	21.1	16.8	11.3	4.2	-3.0
		MCDB	-1.6	4.7	11.9	19.5	23.6	25.6	27.4	25.6	22.1	15.9	8.3	0.2
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.2	14.1	14.8	14.6	13.6	12.1	11.9	13.1	13.5	12.2	12.1
		MCDBR	15.1	16.6	17.9	18.8	18.4	17.2	15.3	14.7	16.5	16.9	15.3	13.3
		MCWBR	10.9	10.2	9.2	7.9	6.5	5.3	4.8	5.2	6.6	8.1	8.7	9.1
	5% WB	MCDBR	14.4	15.1	16.5	15.1	14.2	12.5	11.5	11.2	11.8	13.7	12.6	12.3
		MCWBR	10.6	9.5	8.8	6.8	5.7	4.8	4.8	4.9	5.6	7.0	7.7	8.7
Clear-Sky Solar Irradiance	taub	0.346	0.415	0.443	0.487	0.517	0.581	0.561	0.564	0.515	0.461	0.377	0.348	
	taud	2.114	1.963	1.917	1.870	1.852	1.763	1.869	1.840	1.894	1.951	2.103	2.145	
	Ebn at Noon	790	774	800	793	778	727	738	720	725	716	745	754	
	Edh at Noon	112	150	175	195	203	222	199	199	177	150	112	101	
All-Sky Solar Radiation	RadAvg	2.73	3.56	4.92	5.93	6.59	6.19	5.87	5.31	4.50	3.78	2.87	2.37	
	RadStd	0.21	0.26	0.41	0.38	0.41	0.48	0.27	0.35	0.31	0.28	0.36	0.25	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	-88	-93	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	-1.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page